MILITARY SPECIFICATION

TRANSMITTER, PRESSURE, MANIFOLD, 10-75 INCH MERCURY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of remote indicating manifold pressure transmitter, designated Type MS28005-6.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

PPP-B-601	Boxes, Wood, Cleated Plywood
PPP-B-636	Box, Shipping, Fiberboard

MILITARY

MIL-P-116	Preservation, Method Of
MIL-T-5350	Transmitters, Synchro Operated Aircraft, General Specification For

STANDARDS

MILITARY

MIL-STD-129	Marking For Shipment And Storage

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Oklahoma City Air Logistics Center/MMEDO Tinker AFB OK 73145-5990 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6620
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MIL-STD-2073/1A — DOD Material Procedures For Development And Application Of Packaging Requirements

MS STANDARDS

MS28005 — Transmitter, Pressure, Synchro Aircraft

(Copies of specification, standards, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

American Society For Testing And Materials (ASTM)

ASTM D3951 Packaging, Commercial

(Application for copies should be addressed to: ASTM, 1916 Race St, Philadephia PA 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event between the text of this specification and the reference cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.2 General specification. The requirements of MIL-T-5350 apply as requirements of this specification with the exceptions and additions called out herein. When the two specifications conflict, this specification shall govern.

3.3 Design and construction.

3.3.1 Range. The transmitter synchro shall move through an arc of $345^{\circ} \pm 15$ minutes for the range of 10 to 75 inches Hg.

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3.3.2 Damping. The transmitter shall be sufficiently damped to prevent oscillation of the indicator pointer but shall not introduce an objectionable lag in pointer response.

3.3.3 Stop. A stop shall be provided and shall be so designed as to restrain the primer mover. The stop shall allow the pointer of the test indicator to travel beyond full scale to a point approximately midway between the first and last graduations.

3.3.4 Outline dimensions. The outline dimensions of the transmitter shall conform to MS28005.

3.3.5 Noncorrosive seal. The mechanism of the transmitter unit shall be sealed from the pressure compartment by a metal diaphragm or bellows to prevent vapors from coming in contact with the mechanism.

3.4 Performance. Each transmitter shall be capable of operating one or two indicators without a change in indication of more than 1° of arc.

3.4.1 Performance exceptions. The shock and acceleration tests of MIL-T-5350 shall not be required.

3.4.2 Temperature range. The operational temperature range of the transmitter shall be $-55^{o}C \pm 2^{o}C$ to $+70^{o}C \pm 2^{o}C$.

3.5 Weight. The weight of each transmitter shall not exceed 1.30 pounds.

3.6 Recycled and reclaimed materials. Recycled and reclaimed materials shall be encouraged to the maximum extent possible without jeopardizing the intended end use of the item.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

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4.2 General specification. The transmitter shall be subjected to inspection and tests in accordance with MIL-T-5350 and as specified herein. Errors shall not exceed the applicable tolerances of Tables I or II. The following tests of MIL-T-5350 shall be performed as qualification tests but shall not be required as quality conformance tests.

a. Radio noise suppression

b. Salt fog

4.3 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.5 thru 4.5.2).

b. Quality conformance inspection (see 4.5.3).

4.4 Inspection conditions. Unless otherwise specified, the transmitter shall be tested with the longitudinal axis in a horizontal plane.

TABLE I. Manifold Pressure Transmitter-Scale Error Tolerance.

Test Pressure Inches Hg	Room Temperature	Salt Fog Vibration Overpressure	Low Temperature High Temperature Seasoning Endurance
30	0.20 in. Hg		
25	0.30		
20	0.40		
15	0.40		
10	0.50	0.70 in. Hg	0.80
20	0.40	0.50	0.60
30	0.20	0.30	0.40
35	0.20	0.30	0.40
40	0.20	0.30	0.40
45	0.20	0.30	0.40
50	0.30	0.40	0.50
55	0.30	0.40	0.50
60	0.40	0.50	0.60
65	0.40	0.50	0.60
70	0.50	0.60	0.80
75	0.50	0.60	0.80

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TABLE II. Pointer Error and Friction Loss Tolerances.

Test	Tolerances-Inches Hg
Friction:	
Room Temperature	0.50
Low Temperature	0.75
High Temperature	0.50
Salt fog	0.50
Position Error at 50 in. Indication	0.25
Vibration: Pointer Oscillation	0.75

4.5 First article inspection.

4.5.1 Inspection samples. The first article inspection samples shall consist of three transmitters representative of the production equipment. The samples shall be identified with the manufacturer's part number and such other information as required by the qualifying activity.

4.5.2 Quality conformance inspection. The quality conformance inspection shall consist of all the tests of this specification including the tests of MIL-T-5350 which are applicable to this specification. Unless otherwise specified by the qualifying activity, tests shall be conducted in the order listed.

4.6 Individual inspection. The individual inspection tests shall consist of the individual inspection tests of MIL-T-5350 together with those of this specification. In case of conflict or duplication, MIL-T-5780 shall govern.

4.6.1 Leakage test (nonhermetically sealed only). A pressure of 10 inches Hg absolute shall be applied to the pressure connection of the transmitter. The connecting tubing shall then be pinched off or otherwise completely sealed at a point within 2 inches of the pressure connection. During a period of 1 minute there shall be no indication of leakage.

4.6.2 Damping. A pressure of 75 inches Hg shall be applied to the pressure connection of the transmitter. The pressure shall then be suddenly released and the indicator shall traverse the interval from 70 to 32 inches Hg absolute within a period of 1.5 to 3.0 seconds.

4.7 Sampling plan A. Sampling Plan A tests shall consist of the Sampling Plan A tests of MIL-T-5350 together with those of this specification. In case of conflict or duplication, MIL-T-5780 shall govern.

4.7.1 Vacuum and overpressure. A pressure of 80 inches Hg shall be applied to the transmitter for a period of 10 minutes. The pressure shall then be reduced to 5 inches Hg absolute and maintained for a period of 10 minutes. This pressure shall be released and not more than 5 minutes later the transmitter shall be subjected to the scale error test. Errors shall not exceed tolerances shown on Table I.

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4.7.2 High temperature. The transmitter shall be subjected to a temperature of $70^{\circ} \pm 2^{\circ}C$ for a period of 6 hours. The transmitter shall be electrically connected to an appropriate indicator with power applied during this test. At the end of this period and with the transmitter still at this temperature the transmitter shall be subjected to a scale error and friction test. Errors shall not exceed the tolerances specified in Tables I and II.

4.8 Sampling Plan B. Sampling Plan B shall consist of the Sampling Plan B tests of MIL-T-5350 without duplication.

4.9 Inspection of packaging. Test of methods of preservation and packaging shall be accomplished in accordance with Section 4 of MIL-P-116 to insure compliance with Section 5 of this specification.

5. PACKAGING

5.1 Preservation. Preservation shall be level A, C, or Industrial, IAW MIL-STD-2073/1A, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning. Transmitter shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. Transmitter shall be dried in accordance with process D-4 of MIL-P-116.

5.1.1.3 Preservation application. Preservative shall not be used.

5.1.1.4 Unit packaging. Unless otherwise specified by the contracting activity, each transmitter shall be packaged in quantity unit packs of one each in accordance with Method III of MIL-P-116. Each transmitter shall be placed in a PPP-B-636 Fiberboard container weather resistant, with sufficient cushioning material between bag and unit container of a type, density, and thickness to insure shock transmission does not exceed peak values in G's established for the indicator when completed packs are subjected to the rough handling drop tests of MIL-P-116.

5.1.2 Level C. Each transmitter shall be clean, dry, and individually packaged in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from supply source to the first receiving activity.

5.1.3 Industrial. The Industrial preservation of transmitter shall be in accordance with ASTM D3951.

5.2 Packing. Packing shall be level A, B, C, or Industrial as specified (see 6.2).

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5.2.1 Level A. Transmitter packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-601, Styles A or B, Class overseas, unless otherwise specified by the contracting activity. Insofar as practical, exterior shipping container shall be of uniform shape, size, minimum tare and cube consistent with the protection required.

5.2.2 Level B. Transmitter packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-636, class-weather resistant, unless otherwise specified by the contracting activity. Other requirements as specified in 5.2.1 aplly.

5.2.3 Level C. Packing shall be applied which affords adequate protection during domestic shipment from the supply source to the first receiving activity for immediate use. This level shall conform to applicable carrier rules and regulations.

5.2.4 Industrial. The packaged transmitter shall be packed in accordance with ASTM D3951.

5.3 Marking. In addition to any other markings required by the contract or order (see 6.2), interior and exterior containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The manifold pressure transmitter covered by this specification is intended for use in aircraft to transmit the absolute pressure in the intake manifold of aircraft engines to an applicable indicator.

6.2 Ordering data. Procurement documents should specify.

a. Title, number, and date of this specification.

b. When sampling plan B tests will not be conducted (see MIL-T-5350).

c. Applicable levels of packaging and packing.

6.3 First article. A first article is required for inspection and it shall be tested and approved under the appropriate provisions of 7-104.55 of the Defense Acquisition Regulation. First article tests are described in 4.3. The contracting officer should include specific instructions in all procurement instruments, regarding arrangements for examinations, tests and approval of the first article.

6.4 Subject term (key word listing).
Manifold
Pressure
Transmitter
10-75 Inch mercury
MS28005-6

Custodian:
AIR FORCE -99
NAVY -AS
ARMY -AV

Preparing Activity:
AIR FORCE -71

Project number:
6620-0443

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-T-5780D	TRANSMITTER, PRESSURE, MANIFOLD, 10-75 INCH MERCURY

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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